

AutochipAssignments_zw4892

First, the original autochip_tutorial.ipynb was damaged, wrote a python script to fix it.

```
import re
import os

# 1. 设置文件名 (确保你的 ipynb 文件和这个脚本在同一个文件夹)
input_file = "D:\\advanced_project\\file_recover\\autochip_tutorial.ipynb"
output_file = "D:\\advanced_project\\file_recover\\fixed_autochip.ipynb"

if not os.path.exists(input_file):
    print(f"错误: 找不到文件 {input_file}, 请检查文件名是否正确。")
else:
    with open(input_file, "r", encoding="utf-8") as f:
        content = f.read()

    # 2. 修复逻辑: 利用正则表达式删除 JSON 结构中末尾多余的逗号
    # 修复像 [ "item", ] 这种结构
    fixed_content = re.sub(r',\s*\]', ']', content)
    # 修复像 { "key": "val", } 这种结构
    fixed_content = re.sub(r',\s*\}', '}', fixed_content)

    # 3. 保存修复后的文件
    with open(output_file, "w", encoding="utf-8") as f:
        f.write(fixed_content)

    print(f"✅ 修复完成! 生成了新文件: {output_file}")
    print("现在你可以尝试将 fixed_autochip.ipynb 上传到 Colab 了。")
```

Example1_binary_to_BCD:

Iteration0

Used unproper and vague prompts . So the code's module and ports can't match the name in testbench. Also as the Verilog_loop function only search for testbench named "tb " so the generated code can't be load.

```
user: The design failed to compile. Please fix the module. The output of iverilog
error: Unable to find the root module "tb" in the Verilog source.
```

Iteration1:

Used prompts with more details and give it standard module name and ports name ,also change the name of testbench to load the file.

But at the beginning the testbench has no mismatch information so I can't know the outcome of the simulation.

```
*** /content
  ✓ successfully load handmade file: binary_to_bcd
  start the binary_to_bcd (Trajectory)...
  Iteration: 0
  Model type: ChatGPT
  Model ID: gpt-4o
  Number of responses: 1
  mistake: Simulation output does not contain final mismatch information
```

Iteration2:

Add a mismatch counter in the tb.v file and finally it turns out pass.

```
liverilog -o sim.out binary_to_bcd_tb.v output/iter0/response0/binary_to_bcd.sv
!vvp sim.out

*** Success at input 213!
mismatches: 0
```

Example2_Sequence_detector:

Iteration0:

It turns out that the config.json should be changed in order to match the inside function and the outdir.

Iteration1:

It gave out error as syntax error.

After checking the log, and the code it generate , I found the grammar of the code might be so complicate. So changed the prompts and made it only create in standard Verilog grammar.

```
liverilog -o sim.out sequence_detector_tb.v output/iter0/response0/sequence_detector.sv
!vvp sim.out

-- output/iter0/response0/sequence_detector.sv:9: syntax error
output/iter0/response0/sequence_detector.sv:9: error: invalid module instantiation
output/iter0/response0/sequence_detector.sv:17: syntax error
output/iter0/response0/sequence_detector.sv:17: error: invalid module instantiation
sim.out: Unable to open input file.
```

Iteration 2:

Finally it passed the testbench and comes out with 0 mismatch.

```
--- /content
  [x] successfully load handmade file: sequence_detector
  start the sequence_detector (Trajectory)...
  Iteration: 0
  Model type: ChatGPT
  Model ID: gpt-4o
  Number of responses: 5
  mistake: Simulation output does not contain final mismatch information
```

```
▶ liverlog -o sim.out sequence_detector.th.v output/iter0/response0/sequence_detector.rv
  /vvp sim.out
```

```
--- mismatch: 0
```